

Workflow Documentation

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Things to do:

- [Start pipeline docs](#)
- [Start metrics metadata](#)
- [Expand ArcGIS Portfolio docs](#)

This document is intended to serve as a reference for those of us working on the UNH Sustainability Institute's Split Incentive Tak Force.

1 Data Pipeline

This section will serve as documentation for the data pipeline. The required python libraries include `pandas`, `us`, and `census`. There are some others that are standard.

1.1 Setup

The preamble/defining involves setting the dictionaries `target_states`, `nems_munis`, and `years`. The `us.states` module takes care of the FIPS numbers for each state. Target municipalities (i.e., NEMS members) are taken from a master list; formatting them correctly is a great use for a chatbot. This output needs to be verified, though. The years of interest will depend on the application; however, one should not compare overlapping ACS5 datasets! [Click here to see why.](#)

1.2 Getting ACS5 Data

Currently there are [two different API calls](#) taking place: one for housing age-related data, and one for the rest. This will change.

Query helper

Below are some useful facts about the functions used to interact with the ACS5 API.

```
acs_df_from_raw(  
    all_raw_data,  
    rename_dict=dict)
```

Returns a Pandas DataFrame from raw data dumped out of the ACS5 module of the census package: `census.acs5.state_county_subdivision`.

```
query_states_years_variables(  
    target_states=dict,  
    target_years=list,  
    variables=dict,  
    loud=True)
```

Essentially a wrapper for `census.acs5.state_county_subdivision`. It manually deals with some issues with a few of the municipalities. Some naming conventions aren't matching, so I've manually changed them. This happens despite GEOID being a unique identifier; it may point to a misaligned set of underlying shapefiles. This requires further investigation.

Housing Age

Several different functions exist which are intended to pull different “classes” of ACS5 variables. For instance, *Renter Landscape* (renter share, rent burden), *Housing* (housing age, size), or *Energy* (energy burden, heating fuel). The only real difference currently is between variables related to housing age and those not related to housing age.

1.3 Getting GIS Data

2 ArcGIS Online

The goal of this section is to detail the process of turning **.shp** files derived from **GeoDataFrame** into ArcGIS Instant Apps, for use in Portfolio projects. The system schematic is as follows:

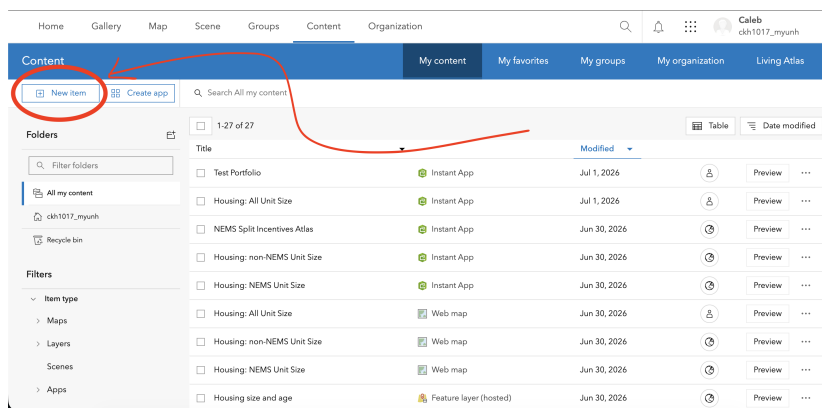
Portfolio ← Instant App ← Web Map ← Feature Layer

IMPORTANT: To do any of this you will need an ArcGIS Online account with publishing permissions. UNH has a license for ArcGIS Online, but we need to contact admin to get permissions. Let me know if you don't have publishing permissions.

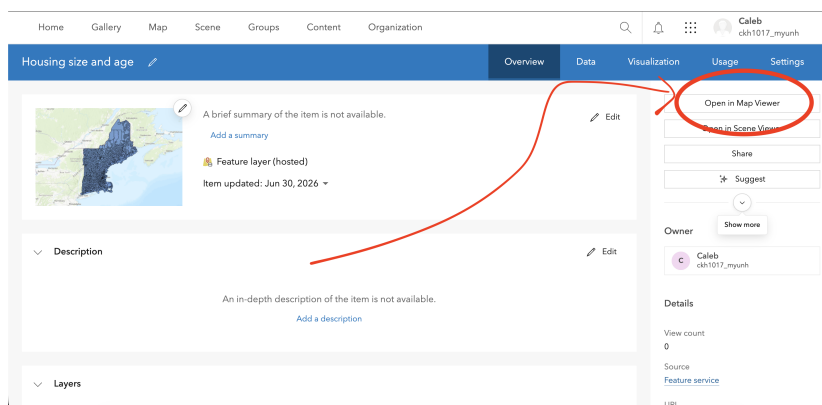
Create an Instant App

A **Portfolio** is a collection of **sections** which link to Web Apps. We will be tying into the efforts of and extending the work of BU's CELT program. With this in mind, our propotypical example of a Portfolio should be their Legacy Transition Atlas, which is linked [here](#).

1. Prepare ShapeFiles. Be sure that the relevant **GeoDataFrame** has been exported and compressed into a **.zip** file. The process of obtaining the shapefiles is outlined in Section 1.
2. Upload the **.zip** file to ArcGIS Online as a **Feature Layer**. Navigate to the **Content** tab of ArcGIS Online, then click **New item**. Click the **Feature layer**, and then the **Upload a file** options. Be sure to choose the upload option that creates a Hosted Feature Layer.¹ Pick a good name, and add a helpful description.



3. Back in the **Content** window, open the Hosted Feature Layer you just created. Choose the option to **Open in Map Viewer** in the upper right.



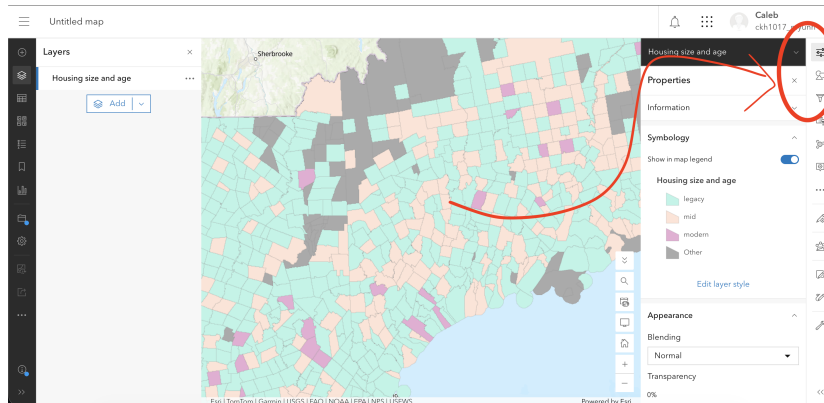
4. Fiddle and Save. The righthand side of the screen has a menu for changing how the map looks. We will almost exclusively tweak the top three options: Properties, Style, and Filters. Here are some style notes which may be standardized:

- 0% transparency

¹This step is not possible without publishing permissions!

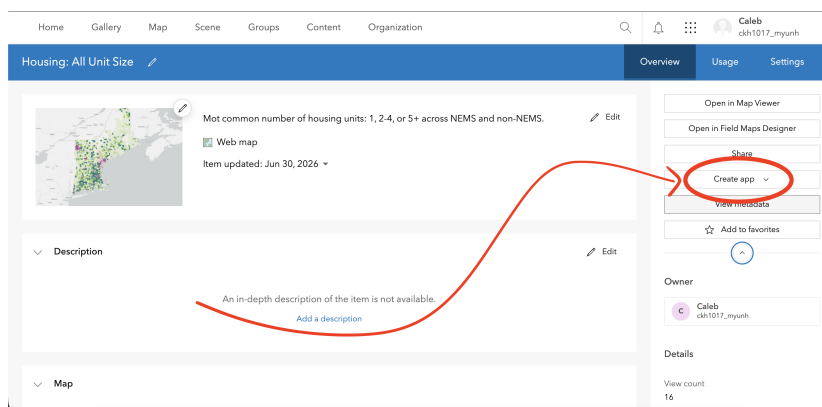
- Continuous variables classed into three quantiles
- NEMS members are colored Purple 1, and non-NEMS are colored Green 5. If colors need to be copied exactly, they can be obtained by opening another map and inspecting the style options.

Now is when the NEMS/non-NEMS filter should be applied, i.e., filter for `nems=1` or `nems=0`.



When you're done, click **Save as** from the lefthand menu and pick the same name you used for the Feature Layer; add a short description as well.

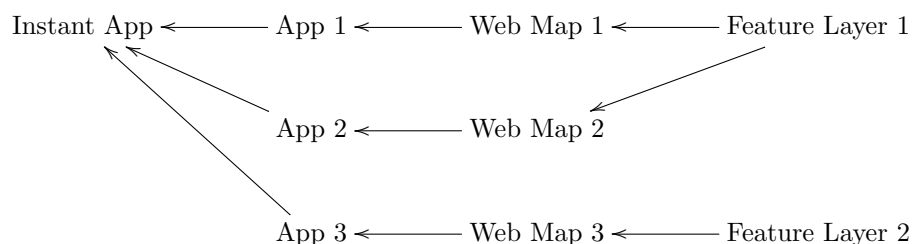
5. In the **Content** window, open Web Map you just created. In the options on the right, click **Create app** and choose **Instant App**. In the page that opens, choose the **Basic (Media Map)** option. Name it the same thing as the previous two. Not much needs to change here.



Thus you have created an Instant App! This app can be shared directly via its URL. Its default visibility is set to Private, meaning only the owner can view it. This can be changed at the current step, or later when publishing a Portfolio.

Create a Portfolio

A portfolio is the housing for possibly several different Instant Apps:



UNSTABLE BELOW

1. Create sections. Choose a base map for each section. There is a map called Base New England owned by me (username `ckh1017_myunh`) which should serve as a nice blank opening screen.

2. Put links to other apps in descriptopn.